DBMS – SALES AND CUSTOMER ANALYTICS SYSTEM

TATA CLiQ E-Commerce Order Management System

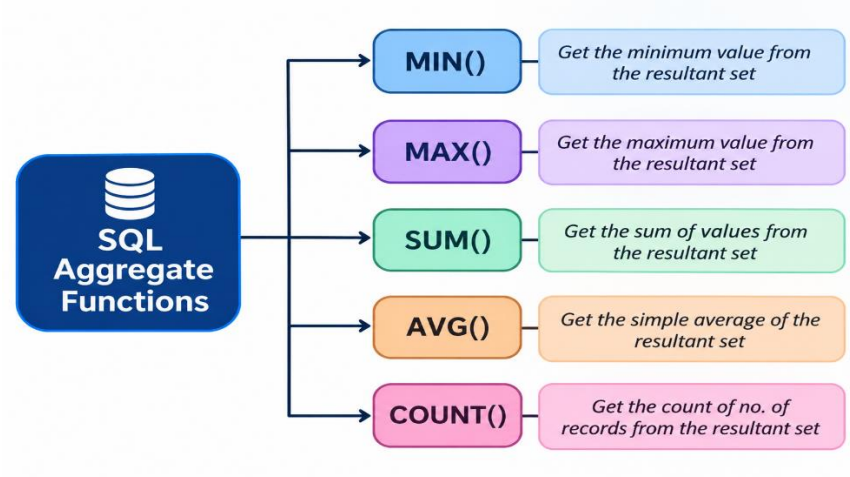
Objectives

1. Apply COUNT(), SUM(), AVG(), MIN(), and MAX().
2. Generate total sales reports.
3. Find top customers based on purchase amount.
4. Identify best-selling products.
5. Perform category-wise sales analysis.

1. Aggregate Functions

Definition

Aggregate functions are SQL functions used to perform calculations on a group of rows and return a single result.



1.1 COUNT() Find the total number of order:

```
SQL> SELECT COUNT(*) AS TOTAL_ORDERS
  2  FROM ORDERS;

TOTAL_ORDERS
------------
           5
```

1.2 SUM() Find the total sales amount:

```
SQL> SELECT SUM(TOTAL_AMOUNT) AS TOTAL_SALES
  2  FROM ORDERS;

TOTAL_SALES
-----------
      14600
```

1.3 AVG() Find the average order amount:

```
SQL> SELECT AVG(TOTAL_AMOUNT) AS AVERAGE_ORDER_AMOUNT
  2  FROM ORDERS;

AVERAGE_ORDER_AMOUNT
--------------------
                2920
```

1.4 MIN() Find the lowest order amount:

```
SQL> SELECT MIN(TOTAL_AMOUNT) AS LOWEST_ORDER_AMOUNT
  2  FROM ORDERS;

LOWEST_ORDER_AMOUNT
-------------------
               1500
```

1.5 MAX() Find the highest order amount:

```
SQL> SELECT MAX(TOTAL_AMOUNT) AS HIGHEST_ORDER_AMOUNT
  2  FROM ORDERS;

HIGHEST_ORDER_AMOUNT
--------------------
                4200
```

2. Generate Total Sales Reports

A sales report can be generated using COUNT(), SUM(), and AVG() functions on the ORDERS table.

```
SQL> SELECT
  2      COUNT(*) AS TOTAL_ORDERS,
  3      SUM(TOTAL_AMOUNT) AS TOTAL_SALES,
  4      AVG(TOTAL_AMOUNT) AS AVERAGE_SALES
  5  FROM ORDERS;

TOTAL_ORDERS TOTAL_SALES AVERAGE_SALES
------------ ----------- -------------
           5       14600          2920
```

2.1 Customer-wise Sales Report

This report displays each customer's total purchase amount and arranges customers according to their purchase amount.

```
SQL> SELECT
  2      C.CUSTOMER_ID,
  3      C.FIRST_NAME || ' ' || C.LAST_NAME AS CUSTOMER_NAME,
  4      SUM(O.TOTAL_AMOUNT) AS TOTAL_PURCHASE
  5  FROM CUSTOMER C
  6  JOIN ORDERS O
  7  ON C.CUSTOMER_ID = O.CUSTOMER_ID
  8  GROUP BY C.CUSTOMER_ID, C.FIRST_NAME, C.LAST_NAME
  9  ORDER BY TOTAL_PURCHASE DESC;

CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------------------
----
TOTAL_PURCHASE
--------------
        101
Thansila Begam
         4200

        104
Divya Rani
         4200

CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------------------
----
TOTAL_PURCHASE
--------------
```

3. Find Top Customers Based on Purchase Amount

The SUM() function is combined with GROUP BY and ORDER BY to identify customers with higher purchase amounts.

```
SQL> SELECT
  2      C.CUSTOMER_ID,
  3      C.FIRST_NAME || ' ' || C.LAST_NAME AS CUSTOMER_NAME,
  4      SUM(O.TOTAL_AMOUNT) AS TOTAL_PURCHASE
  5  FROM CUSTOMER C
  6  JOIN ORDERS O
  7  ON C.CUSTOMER_ID = O.CUSTOMER_ID
  8  GROUP BY C.CUSTOMER_ID, C.FIRST_NAME, C.LAST_NAME
  9  ORDER BY TOTAL_PURCHASE DESC;

CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------------------------
----
TOTAL_PURCHASE
--------------
        101
Thansila Begam
          4200

        104
Divya Rani
          4200

CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------------------------
----
TOTAL_PURCHASE
--------------

        103
```

4. Identify Best-Selling Products

The ORDER_DETAILS table contains the quantity of each product ordered. SUM(QUANTITY) finds the total quantity sold for each product.

```
SQL> SELECT
  2      P.PRODUCT_ID,
  3      P.PRODUCT_NAME,
  4      SUM(OD.QUANTITY) AS TOTAL_QUANTITY_SOLD
  5  FROM PRODUCT P
  6  JOIN ORDER_DETAILS OD
  7  ON P.PRODUCT_ID = OD.PRODUCT_ID
  8  GROUP BY P.PRODUCT_ID, P.PRODUCT_NAME
  9  ORDER BY TOTAL_QUANTITY_SOLD DESC;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------
----
TOTAL_QUANTITY_SOLD
-------------------
       201
Laptop
                  3

       203
Mobile
                  2

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------
----
TOTAL_QUANTITY_SOLD
-------------------

       204
Grocery
                  2

       206
Headphones
```

This query connects PRODUCT and ORDER_DETAILS, groups records by product, calculates total quantity sold, and displays products from highest to lowest quantity sold.

4.1 Product Sales Amount

This query calculates both the total quantity sold and the sales amount generated by each product.

```
SQL> SELECT
  2      P.PRODUCT_ID,
  3      P.PRODUCT_NAME,
  4      SUM(OD.QUANTITY) AS TOTAL_QUANTITY_SOLD,
  5      SUM(OD.QUANTITY * P.PRICE) AS TOTAL_SALES_AMOUNT
  6  FROM PRODUCT P
  7  JOIN ORDER_DETAILS OD
  8  ON P.PRODUCT_ID = OD.PRODUCT_ID
  9  GROUP BY P.PRODUCT_ID, P.PRODUCT_NAME
 10  ORDER BY TOTAL_SALES_AMOUNT DESC;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------
----
TOTAL_QUANTITY_SOLD TOTAL_SALES_AMOUNT
------------------- ------------------
       201
Laptop
                  3             150000

       203
Mobile
                  2              40000

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------
----
TOTAL_QUANTITY_SOLD TOTAL_SALES_AMOUNT
------------------- ------------------

       206
Headphones
                  2               4000

       205
```

5. Category-wise Sales Analysis

Category-wise analysis helps determine the sales generated by each product category.

```
SQL> SELECT
  2      C.CATEGORY_ID,
  3      C.CATEGORY_NAME,
  4      SUM(OD.QUANTITY * P.PRICE) AS TOTAL_SALES
  5  FROM CATEGORY C
  6  JOIN PRODUCT P
  7  ON C.CATEGORY_ID = P.CATEGORY_ID
  8  JOIN ORDER_DETAILS OD
  9  ON P.PRODUCT_ID = OD.PRODUCT_ID
 10  GROUP BY C.CATEGORY_ID, C.CATEGORY_NAME
 11  ORDER BY TOTAL_SALES DESC;

CATEGORY_ID CATEGORY_NAME                                      TOTAL_SALES
----------- -------------------------------------------------- -----------
          1 Electronics                                             150000
          3 Laptops                                                  40000
          6 Beauty                                                    4000
          5 Footwear                                                  2500
          2 Mobiles                                                   1500
          4 Fashion                                                   1000

6 rows selected.
```

6. Combined Business Sales Report

A complete sales report combines customer, order, product, and category information.

```
SQL> SELECT C.CUSTOMER_ID, C.FIRST_NAME || ' ' || C.LAST_NAME AS CUSTOMER_NA
ME, O.ORDER_ID, O.ORDER_DATE, P.PRODUCT_NAME, CAT.CATEGORY_NAME, OD.QUANTITY
, P.PRICE, (OD.QUANTITY * P.PRICE) AS SALES_AMOUNT, O.ORDER_STATUS
  2  FROM CUSTOMER C
  3  JOIN ORDERS O ON C.CUSTOMER_ID = O.CUSTOMER_ID
  4  JOIN ORDER_DETAILS OD ON O.ORDER_ID = OD.ORDER_ID
  5  JOIN PRODUCT P ON OD.PRODUCT_ID = P.PRODUCT_ID
  6  JOIN CATEGORY CAT ON P.CATEGORY_ID = CAT.CATEGORY_ID
  7  ORDER BY O.ORDER_DATE DESC;

CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------------------------------------
----
  ORDER_ID ORDER_DAT
---------- ---------
PRODUCT_NAME
--------------------------------------------------------------------------------
----
CATEGORY_NAME                                      QUANTITY      PRICE
-------------------------------------------------- ---------- ----------
SALES_AMOUNT ORDER_STATUS
------------ --------------------
         102

CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------------------------------------
----
  ORDER_ID ORDER_DAT
---------- ---------
PRODUCT_NAME
```